Scale Factor: Perimeter and Area

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lessons 2–3 — Scale Factor: Perimeter and Area	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.GR.1.5

2. Standards

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must. Find the perimeter and area of a scale copy from the original and k (perimeter × k, area × k^2), and find k from two areas when the ratio is a perfect square. (MA.7.GR.1.5, clarification 2.)

Must not. Area ratios that are not perfect squares (an approximate root); the area of a figure the class has no formula for; area and perimeter in different units (5.04 and 5.08–09).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes II: the same copy as a grid picture, as 15 → 60, and as $k^2 = 4$
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote on k against k^2
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	boards 1–2: the same k on the perimeter, then the area, and the room is asked why the second answer is bigger

4. Learning Target and Essential Question

Learning target. I can find the perimeter and the area of a scale copy from the original and k, and find k from two areas.

Essential question. Every side of a rectangle is doubled. Why is its area four times as big, not twice?

Building on. 5.01 — k = copy ÷ original; 4.05 — square roots of perfect squares; Grade 6 — perimeter and area of a rectangle.

Working toward. 5.04 — floor plans, where area has to be found in real units; 5.08–09 — area converts by the square of the length ratio.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
A rectangle's sides are tripled. Its perimeter is multiplied by 3 and its area by 9. Why are the two numbers different?	DOK 3
Two squares have areas 25 and 100. What is k, and how do you know without finding a side?	DOK 2
Could a copy have the same perimeter as its original and a different area? Explain.	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A figure with perimeter 14 cm is scaled by k = 2. What is the copy's perimeter?	***28 cm***	56 — perimeter × k^2 [Notes I: a perimeter is lengths, × k]; 16 — added k [7.GR.1.5: the additive error — Math Nation's named misconception]
2	A figure with area 12 cm^2 is scaled by k = 2. What is the copy's area?	***48 cm^2***	24 — area × k [B1G-M 7.GR.1.5 named misconception: scaling area like length]
3	A figure with area 48 cm^2 is scaled by k = ½. What is the copy's area?	***12 cm^2***	24 — area × k [7.GR.1.5: area × k^2]; 96 — divided by ½ as if k were 2 [5.01: k under 1 shrinks]
4	A figure with area 10 cm^2 is scaled by k = 3. What is the copy's area?	***A — 90 cm^2***	(B) area × 2k: 'two lengths, so twice k' [Notes II: k × k, not k + k]; (C) area × k: the guide's named misconception [B1G-M 7.GR.1.5: scaling area like length]; (D) 10 + 3: the additive error [7.GR.1.5: Math Nation's named misconception]
5	A figure with area 8 cm^2 is scaled by k = 1.5. What is the copy's area?	***18 cm^2***	12 — area × k [7.GR.1.5]; 24 — area × 2k [Notes II: k × k]
6	A figure's perimeter is 20 cm; its copy's perimeter is 60 cm. What is the area ratio, copy to original?	***D — 9***	(A) 2k: 'area has two dimensions, so double it' [Notes II]; (B) the perimeter ratio handed in as the area ratio [Notes I and II: perimeter gives k, area is k^2]; (C) 60 − 20: the additive error [7.GR.1.5: Math Nation's named misconception]
7	A triangle with sides 6, 8 and 10 cm is scaled by k = ½. Its area is 24 cm^2. Find the copy's perimeter and area.	***P = 12 cm, A = 6 cm^2***	P 12, A 12 — the area halved like a length [7.GR.1.5]; P 6, A 6 — both quartered
8	A 4 in by 6 in photo, taken 3 years ago, is enlarged at k = 2.5. What is the enlargement's area?	***150 in^2***	60 — area × k [7.GR.1.5]; 120 — 24 × 5, k doubled [Notes II]
9	Jo says that doubling every side of a rectangle doubles its area. Explain why not, with a counterexample.	***No: 2 by 7 doubled is 4 by 14, area 14 → 56 — × 4, as both lengths doubled.***	'Yes, because every side doubles' — scaling area like length [B1G-M 7.GR.1.5]; 'No, it is k^{2t} with no numbers — the rule without the reason

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Two lines stay on the board all period: perimeter × k, area × k × k. Every board is answered by pointing at one of them before multiplying. The grid picture of Notes II stays up beside them.

ELL. **Perimeter** is the distance around; **area** is the squares inside. Say "around" and "inside" with each board, and the word **squared** means times itself.

Enrichment. Board 7 finishes early for some: ask them what k does to a triangle's angles (nothing) and to its height (× k), and why the area rule still holds when there is no length × width.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: a sentence that area is two lengths multiplied, so it is multiplied by k twice, with a counterexample's numbers (a 2 by 7 at k = 2: 14 → 56, not 28). A board that only says 'because k^{2}' has named the rule, not the reason.

Vocabulary. **k squared**; **perimeter ratio**; **area ratio**

Materials. Whiteboards and markers. A sheet of grid paper under the document camera for Notes II's four-copies picture, drawn live if the slide's grid is not enough. Calculators are allowed on everything and printed on nothing.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Perimeter and area: changes in scale — ZC6; Scale drawings: find the scale factor or missing side — XHM.